

Benchmarking Deep-Reinforcement-Learning Anti-Jamming Against Cryptographic Frequency Hopping for GPS-Denied Drone Positioning

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Abstract—Drones increasingly operate where the Global Positioning System (GPS) is jammed or spoofed. Time-Difference-of-Arrival (TDOA) multilateration from ground beacons is a GPS-independent alternative, but only as reliable as the radio channel carrying its signals. We reproduce a deep-Q-network (DQN) anti-jamming model and benchmark it against training-free cryptographic frequency hopping and spread-spectrum baselines under one simulator, a ten-class jammer taxonomy from narrowband to wideband barrage, and a common metric suite, with fixed seeds and released datasets. The maximum-likelihood TDOA estimator attains the Cramér–Rao bound in line-of-sight from 10 dB upward, and RANSAC cuts non-line-of-sight median error by up to $20\times$. The learned model beats cryptographic hopping at small channel counts (avoidance 0.91 vs 0.70 at $M=10$) but fails as channels scale: at $M=1000$ its avoidance is 0.00 even with the 15,000 training updates that bring it to 0.90 at $M=10$. In a representative run, training collapses its policy onto one channel that an adaptive jammer tracks, whereas the same network untrained avoids 0.72 of slots. Cryptographic hopping needs no training, holds its avoidance as the band widens, and reaches ≈ 1.00 against an intelligent jammer that defeats predictable hopping. A valid signature does not stop a delayed replay; an authenticated-timing layer rejects every simulated spoofed and replayed packet, provided the drone can predict each beacon’s time of flight to within nanoseconds. Cryptographic hopping costs an estimated 13 pJ per decision, roughly 700,000 times less than deep-learning inference. We release the code and data.

Index Terms—GPS-denied positioning, TDOA multilateration, cryptographic frequency hopping, deep reinforcement learning, anti-jamming, replay attacks, reproducible benchmark.

I. INTRODUCTION

UNMANNED aerial vehicles now perform safety-critical roles, inspection, delivery, search-and-rescue, and defense, that demand reliable positioning even when an adversary is actively denying it. GPS signals are weak and unauthenticated, so low-cost jammers can deny them [18] and spoofers can inject a false location [24], [25], [19]. A robust, GPS-independent capability is therefore essential. TDOA multilateration provides one: a drone measures the differences in arrival time of signals from beacons at known positions and solves a hyperbolic system for its own location, requiring neither GPS nor an on-board clock [6], [8], [30].

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The labeled spectrum-occupancy and TDOA datasets, the simulator, and all scripts needed to reproduce every figure and table in this paper are available at <https://github.com/Ahmad-prog/secure-tdoa-benchmark> under the MIT license.

Its resilience, however, is only as strong as the radio channel carrying the positioning signals. If those signals are jammed no fix is possible, if a beacon is spoofed the fix is wrong, and if a genuine signal is recorded and replayed with a delay the timing that TDOA relies upon is silently corrupted. Securing the channel is thus inseparable from securing the position.

Two communities offer competing answers. Academic work increasingly applies deep learning to detect jamming and adaptively hop to clean channels [11], [12], [13], [31]. Fielded systems that must survive real jamming instead use cryptographically-driven frequency hopping, in which a shared secret key generates an unpredictable hop sequence [1], [2], [17]. The learned approach is data-driven, adaptive, and avoidance-based; the cryptographic approach is proven, training-free, and unpredictability-based. To our knowledge, no prior work has placed the two side by side, under an identical environment, for drone positioning, nor used cryptographic hopping to secure a TDOA position estimate against the full jam/spoof/replay threat. We therefore ask three questions. Under identical conditions, when does a learned anti-jammer outperform cryptographic hopping? Why does it stop doing so as the band widens? And can the channel that carries the positioning signals also protect the position estimate itself?

A. Problem Statement

Let the drone position be $p \in \mathbb{R}^3$ and let beacons $s_0, \dots, s_{N-1} \in \mathbb{R}^3$ be known. With $r_i = \|p - s_i\|$, the measured range differences relative to reference beacon 0 are

$$d_i = (r_i - r_0) + n_i, \quad i = 1, \dots, N-1, \quad (1)$$

where n_i is estimation noise; the position estimate is the maximum-likelihood solution

$$\hat{p} = \arg \min_p (d - h(p))^\top \Sigma^{-1} (d - h(p)), \quad (2)$$

with $h_i(p) = \|p - s_i\| - \|p - s_0\|$. The positioning signals are carried on one of M channels chosen each slot t by a hopping policy c_t ; a jammer denies a set $J_t \subseteq \{0, \dots, M-1\}$. The security objective is to maximize the jamming-avoidance rate

$$A = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[c_t \notin J_t] \quad (3)$$

while keeping the estimate (2) authentic under spoofing and replay, and while remaining tractable as M grows toward the 2,320-channel configuration of fielded frequency-hopping radios [17].

B. Contributions

- 1) An open, reproducible benchmark with released labeled spectrum-occupancy and TDOA datasets, a ten-class jammer taxonomy spanning narrowband through wideband barrage, and a single simulator hosting every method so that the hopping strategy is the only variable that changes.
- 2) To our knowledge, the first controlled head-to-head of cryptographic hopping against a reproduced DQN anti-jamming model under one environment and metric suite.
- 3) An empirical model-scaling characterization under two training regimes, equal wall clock and equal gradient steps, separating degradation caused by a compute budget from degradation intrinsic to the learning problem, together with a diagnosis of that intrinsic failure: at large channel counts, training collapses the learned policy onto a single trackable channel before it has seen enough samples per channel to learn.
- 4) An authenticated-timing layer that addresses the replay/meaconing gap left open by hopping and by message signatures alike, integrated with rather than replacing an existing protocol stack, together with the timing and positioning accuracy it requires of the drone.
- 5) A per-decision cost and energy analysis, and an explicit account of the operational limits of the cryptographic approach.

The remainder of the paper is organized as follows. Section II reviews related work; Section III defines the system, hopping strategies, and threat model; Section IV gives the learned model, key management, synchronization, and the authenticated-timing layer; Section V describes the experimental setup; Section VI reports results; Section VII discusses boundary conditions and limitations; Section VIII concludes.

II. RELATED WORK

A. TDOA positioning for GPS-denied UAVs

Hyperbolic multilateration from arrival-time differences is a mature localization principle: the Chan–Ho closed form [6] gives a near-maximum-likelihood estimate, Foy’s Taylor-series method [8] refines it, and GCC-PHAT [7] estimates the delays; the CRLB bounds the achievable accuracy [27], [29], [30], and recursive filters track the trajectory [26]. Recent work applies TDOA to UAVs and characterizes altitude, bandwidth, and NLOS bias [9], [10], [15]. *Limitation.* This line assumes a benign channel and does not secure the positioning waveform against an active jamming, spoofing, or replay adversary.

B. Deep-learning anti-jamming

Deep reinforcement learning and convolutional models learn to sense the spectrum and hop to clean channels [11], [12], [13], [31], building on DQN control [35] and learned RF signal processing [33], [34]. *Limitation.* A learned policy needs labeled data and training, is predictable to an intelligent adversary, and its action space grows with the channel count, so scalability to wide-band systems is unaddressed. We reproduce the DQN with coarse-grained spectrum prediction

of Zhang et al. [11] as the learned arm of the comparison, since it specifies its ten-channel environment and jammer types completely enough to re-implement.

C. Cryptographic and spread-spectrum hopping

Cryptographic frequency hopping and spread spectrum are decades-old, proven anti-jamming techniques [1], [4], [5], [32]; uncoordinated and keyed spread spectrum provide jamming resistance and key establishment [2], [16], and follower-jamming limits are known [3]. *Limitation.* These are studied as communication primitives; none is benchmarked against a learned model under one environment, nor applied to secure a TDOA *position* estimate. Cryptographic hopping is adopted here unchanged, as a proven reference point rather than a contribution.

D. GPS spoofing and meaconing

Spoofing and meaconing (delayed replay) are well-documented GNSS threats [24], [25], and machine learning has been used to detect spoofing on UAVs [14]; distance-bounding [23] and message authentication [21] are relevant primitives. *Limitation.* A valid content signature does not protect arrival-time integrity, so replay can corrupt TDOA timing without forging anything, a gap not closed by prior work.

III. SYSTEM AND THREAT MODEL

A. Range noise and the CRLB

The per-range estimation standard deviation follows the time-of-arrival bound [27]

$$\sigma_r = c \sigma_\tau, \quad \sigma_\tau = \frac{1}{2\sqrt{2}\pi\sqrt{\text{SNR}}\beta}, \quad \beta = \frac{B}{\sqrt{3}}, \quad (4)$$

with c the speed of light, B the bandwidth, and β the RMS bandwidth. The range-difference covariance is $\Sigma = \sigma_r^2(I + \mathbf{1}\mathbf{1}^\top)$, and with Jacobian $G_i = u_i - u_0$ (u_k the unit vector to beacon k) the position CRLB is

$$\text{CRLB}(p) = \sqrt{\text{tr}[(G^\top \Sigma^{-1} G)^{-1]}}, \quad \eta = \frac{\text{CRLB}}{\text{RMSE}}. \quad (5)$$

The estimator solves (2) by a Chan–Ho initializer [6] followed by damped (Levenberg–Marquardt) Taylor refinement [8]; under NLOS a RANSAC consensus over beacon subsets [28], [38] rejects biased beacons.

B. Hopping strategies

Both ends derive the channel from an AES-128 keystream [20], [22] under the shared key κ and a synchronized counter,

$$c_t = (\text{AES}_\kappa(t) \bmod 2^{32}) \bmod M, \quad (6)$$

which is statistically uniform and unpredictable to any adversary lacking κ . Direct-sequence spread spectrum (DSSS) and uncoordinated spread spectrum (USS) [2] provide the spread-spectrum baselines, and a public pseudo-random hopper and a fixed channel model keyless, predictable strategies. DSSS

is modeled at the link level by its processing-gain margin: it delivers full throughput while the jammed fraction of the band stays below a tolerance of 0.34, and degrades beyond it. The tolerance is a modeling parameter, not a measured receiver figure.

C. Jammer taxonomy and threat model

Ten jammer classes span three tiers. Tier 1 is non-adaptive (constant, random channel, pulsed, and wideband barrage); Tier 2 is patterned (sweep, comb, partial-band); Tier 3 is intelligent (reactive follower, predictive learning jammer, smart partial-band). The partial-band and smart partial-band jammers each deny a contiguous 30% of the band; the smart variant centers that block on the transmitter's most-used channel. The barrage class denies a fresh random 70% of the band each slot. That fraction is set above the DSSS tolerance so that the taxonomy spans both sides of the processing-gain margin: barrage is the only class that exceeds it, and every narrowband class sits below it. The adversary may additionally inject forged beacon packets (spoofing) or capture and re-transmit genuine packets with an added delay (meaconing replay), the latter corrupting arrival-time integrity without forging any content.

IV. METHOD

A. Learned anti-jamming model

The DQN observes a sliding window of the per-channel occupancy, $s_t \in \{0, 1\}^{W \times M}$, and estimates $Q(s_t, a; \theta)$ with one output per channel; it acts greedily $c_t = \arg \max_a Q$ and is trained on the immediate reward $r_t = \mathbf{1}[c_t \notin J_t]$ (one-step / coarse-grained spectrum prediction, $\gamma=0$) by minimizing $(Q(s_t, c_t; \theta) - r_t)^2$. The output layer has M units, so the parameter count and per-decision cost grow as $O(M)$. Exploration is ϵ -greedy, with ϵ decaying by a factor of 0.9995 per step from 1 to a floor of 0.05. Every 2000 updates the greedy policy is evaluated over one 500-slot episode; a run converges at the first evaluation that reaches 0.90 avoidance.

B. Key management

The hop sequence in (6) is only as available as its key. The baseline threat model assumes pre-shared, pre-provisioned keys loaded before the mission, which is standard practice for keyed frequency hopping and analogous to crypto-fill in fielded radios; no over-the-air exchange is then required. Where a key must be established without a pre-shared secret under active jamming, jamming-resistant key establishment via uncoordinated frequency hopping [16] applies, and authenticated Diffie–Hellman covers a secure pre-flight channel. Rekeying is driven by the same counter, yielding fresh subkeys per epoch. The security of the scheme rests on key secrecy, which we state as an explicit assumption and revisit in Section VII.

C. Hop synchronization

Correct de-hopping requires beacon and drone to agree on the current hop index. Synchronization is ensured at three levels. First, TDOA already requires the beacons to share a common, disciplined timebase, since they are fixed infrastructure at known coordinates, so the hop clock reuses that timebase and introduces no new synchronization burden. Second, the drone acquires initial synchronization through an acquisition/sync-word (late-entry) sequence and thereafter advances its counter locally. Third, periodic sync-refresh slots and a tolerance window absorb clock drift and time-of-flight offset. The tolerance is quantified in Section VI as a function of hop rate.

D. Authenticated timing

Cryptographic hopping resists predictive jamming (its sequence is unpredictable) and spoofing-by-injection (wrong channel and no valid tag) but not replay: an attacker re-transmits a genuine packet with an added delay, corrupting the TDOA timing without forging anything. The layer in Algorithm 1 composes three standard primitives, an HMAC [21], a freshness nonce, and a time-of-flight plausibility test in the manner of distance bounding [23], for the specific purpose of protecting a *time-of-arrival* measurement rather than message content. It is deployed alongside whatever authentication an existing protocol stack already provides and replaces none of it; Section VI quantifies why the stack's own signature is not sufficient on its own.

The layer is demanding in two ways. First, the time-of-flight test needs the drone to predict each beacon's geometric delay δ , which depends on its own position, and needs its clock aligned with the beacon timebase, both to within τ . At 20 dB, τ is 5.8 ns, or 1.75 m of range. The drone's own TDOA fix is too coarse to supply δ at that tolerance (7.8 m RMSE at 20 dB, Section VI). A deployment therefore needs an independent prediction, for example a tracking filter with inertial aiding, or a tolerance widened to its position uncertainty, which in turn admits replays delayed by less than that uncertainty. Second, the freshness nonce requires the drone to send a challenge, so the layer gives up the passive, receive-only operation TDOA otherwise allows. The nonce defeats a replay of a previously captured packet; a real-time rebroadcast carries the current nonce and is stopped only by the time-of-flight test.

E. Evaluation metrics

The jamming-avoidance rate A is given by (3); positioning error is $\text{RMSE} = \sqrt{\frac{1}{n} \sum_i \|\hat{p}_i - p_i\|^2}$ with efficiency η from (5); attack success is $\rho = \frac{1}{K} \sum_k \mathbf{1}[\text{attack}_k \text{ accepted} \wedge \text{harmful}]$, where a replay is harmful when its added delay exceeds τ , so that it corrupts the range measurement; normalized throughput is the per-slot delivered fraction; and per-decision energy is $E = \text{MACs} \times e_{\text{MAC}}$ for the DQN, with $e_{\text{MAC}} = 3.7 \text{ pJ}$ at the 45 nm node [39], versus one AES-128 block for cryptographic hopping, taken as 13 pJ for a hardware AES engine (about 0.8 pJ per byte).

Algorithm 1 Authenticated-timing verification at the drone

Require: packet (id, T_{tx} , nonce, tag), key κ , arrival T_{rx} , geometric delay δ , issued nonce ν , tolerance $\tau = 3\sigma_r/c$

Ensure: accept / reject

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1: if tag  $\neq$  HMAC $_{\kappa}$ (id ||  $T_{tx}$  || nonce) then
2:   return reject  $\triangleright$  forged or altered
3: end if
4: if nonce  $\neq \nu$  then
5:   return reject  $\triangleright$  stale / replayed
6: end if
7: if  $|T_{rx} - (T_{tx} + \delta)| > \tau$  then
8:   return reject  $\triangleright$  delayed re-transmission
9: end if
10: return accept

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V. EXPERIMENTAL SETUP

The simulator is self-contained and CPU-only (Python, NumPy/SciPy, the cryptography library); the DQN is a from-scratch NumPy network (window $W=16$, hidden 64, Adam with learning rate 10^{-3} , replay 2000, batch 32), which makes its $O(M)$ scaling cost a measured quantity rather than an assertion. Positioning uses $N=8$ ground beacons on a 1 km hexagon with mast-height diversity and a drone at ~ 300 m. Spectrum episodes are 500 slots; anti-jam results average 30 seeds, the equal-wall-clock scaling arm 10 seeds, the equal-gradient-step arm 3 seeds, the significance tests an independent replicate of 8 seeds, and the policy-collapse probe one representative seed; every run is reproducible from a fixed seed. Channel scales are $\{10, 100, 1000, 2320\}$, the last being the SINCGARS 30–88 MHz/25 kHz configuration [17]. The labeled spectrum-occupancy dataset (Fig. 1) is released with the code.

VI. RESULTS

A. Positioning accuracy against the CRLB

The maximum-likelihood estimator attains the CRLB in LOS from 10 dB upward, with efficiency η between 0.97 and 1.06 (RMSE 7.8 m against a CRLB of 8.2 m at 20 dB). Values slightly above 1 are within the Monte-Carlo error of 500 trials, about 3% relative. Below 10 dB the efficiency falls to 0.91, the usual threshold behavior of the maximum-likelihood estimator at low SNR.

These results use the analytic range-noise model (4). A signal-level GCC-PHAT check does not reproduce that model across SNR: its empirical error stays between 0.43 and 0.85 m from 0 to 30 dB, whereas the analytic σ_r falls from 5.84 to 0.18 m, and the two agree only near 20 dB (ratio 0.79). The positioning results therefore describe the analytic noise model, not a validated signal-level receiver.

NLOS error is heavy-tailed, so we report the median: RANSAC mitigation reduces it from 133 m to 33 m at 20 dB and from 148 m to 7.4 m at 30 dB (Table I, Fig. 2).

B. Anti-jamming across the taxonomy

Against the ten-class taxonomy, cryptographic hopping avoids collisions with a rate that rises with M (mean avoidance

TABLE I
TDOA POSITIONING (METERS). LOS REACHES THE CRLB; RANSAC MITIGATION CUTS NLOS MEDIAN ERROR UP TO $20\times$.

SNR (dB)	LOS RMSE	CRLB	NLOS median	NLOS + mitig.
10	26.8	26.6	138.4	67.8
20	7.8	8.2	133.2	33.3
30	2.7	2.6	147.7	7.4

over the ten classes $0.74 \rightarrow 0.87$) and, crucially, resists the intelligent adversary: against the learning jammer its avoidance is $0.91 \rightarrow 1.00$ while predictable public-random and fixed hopping collapse to 0.00 (Fig. 3). This is the operational meaning of cryptographic unpredictability, and it is what a keyless pseudo-random hopper cannot provide.

The taxonomy also bounds the spread-spectrum baseline. DSSS is unaffected by every narrowband class, because none of them exceeds its processing-gain margin, but wideband barrage denies 70% of the band at once, drops DSSS throughput to 0.45, and takes its avoidance to 0.00. No hopping strategy fares better in that column: with 70% of the band denied, every hopper, cryptographic or otherwise, is reduced to 0.30. Barrage is therefore the one adversary in the study that defeats both families, which is the expected consequence of spreading jamming power across the entire band rather than concentrating it.

C. Learned versus cryptographic hopping at scale

The DQN, when it can be trained, beats cryptographic hopping on structured adversaries, since it learns to occupy clean channels: 0.91 against 0.70 at $M=10$. The advantage does not survive scaling. Under a fixed 20 s compute budget, avoidance falls to 0.07 at $M=1000$, while training throughput collapses from 1,655 to 5 steps/s and the parameter count grows from 1.1×10^4 to 2.5×10^6 (Table II, Figs. 4 and 5). Cryptographic hopping stays at 0.70 at every M , because this jammer always covers 30% of the band, and it needs no training.

A fixed wall-clock budget starves the wide-band models by construction, so we repeat the experiment with the budget removed and the number of gradient updates held at 15,000 for every M . If the degradation were only a budget effect, this arm would recover it. It does not: equal training gives 0.00 at $M=1000$ and 0.00 at $M=2320$ on every seed, and no model at either size reaches 0.90 avoidance within the limit, whereas at $M=10$ and $M=100$ that level is reached after a mean of 3,333 and 8,667 updates over three seeds (Fig. 6); evaluations run every 2,000 updates, so each seed's value is a multiple of 2,000. The equal-wall-clock value at $M=2320$ (0.16 ± 0.27) is not a recovery either. At 5 steps/s that arm completes only about 108 updates, and Table III shows that at $M=1000$ as few as 220 updates already concentrate the policy on one channel. Its wide spread across seeds is consistent with some runs having collapsed and others not yet.

Why more training does not help: Table III probes the greedy policy the DQN would deploy at three points in train-

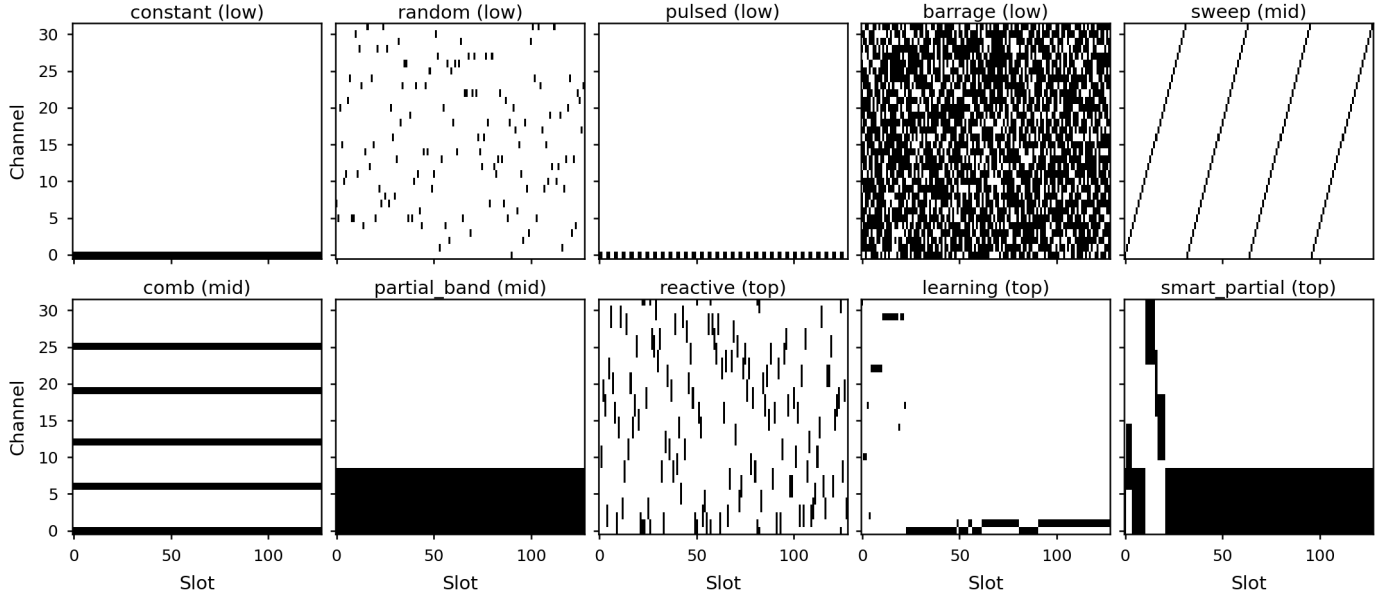


Fig. 1. Released spectrum-occupancy dataset: per-slot channel state (jammed in black) for the ten jammer classes; over a sliding window this is the DQN input.

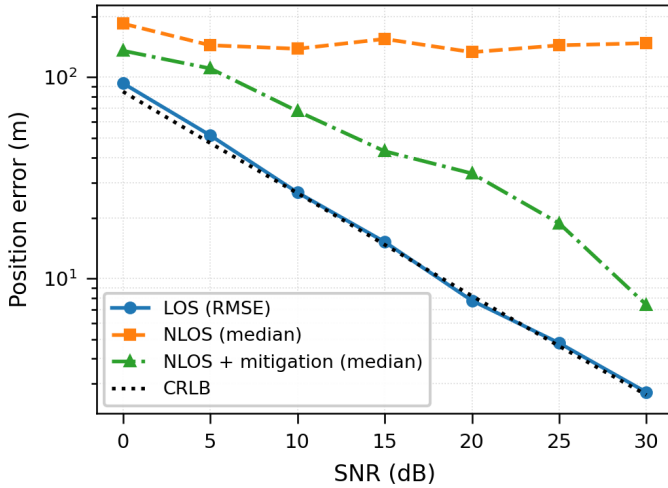


Fig. 2. Positioning error vs SNR. LOS RMSE follows the CRLB from 10 dB upward; NLOS mitigation (median) approaches it as SNR grows.

ing, for one representative seed; the zero variance of the equal-steps arm across seeds at $M=1000$ is consistent with it. At $M=100$ training eventually diversifies the policy: after 3000 updates it uses more distinct channels than at initialization, its most-used channel takes a smaller share, and avoidance reaches 0.90. At $M=1000$ the reverse happens. The untrained network avoids 0.72 of slots, close to cryptographic hopping, because the argmax of a randomly initialized network shifts with its input and so behaves as a pseudo-random hopper. After 220 updates, about what the 20 s budget allows at $M=1000$, 93% of its choices go to a single channel, and after 3000 updates 100%. The smart partial-band jammer centers its block on the transmitter's busiest channel, so it covers that channel in every slot.

The cause is sample complexity. Only the value of the

TABLE II
DEEP LEARNING AT SCALE AGAINST THE SMART PARTIAL-BAND JAMMER, WHICH COVERS 30% OF THE BAND, UNDER AN EQUAL-WALL-CLOCK BUDGET (10 SEEDS) AND EQUAL GRADIENT STEPS (3 SEEDS).

M	DQN (20 s)	DQN (equal steps)	Crypto	Train steps/s
10	0.91	0.90	0.70	1,655
100	0.84	0.83	0.70	181
1,000	0.07	0.00	0.70	11
2,320	0.16	0.00	0.70	5

TABLE III
THE GREEDY POLICY THE DQN WOULD DEPLOY, PROBED DURING TRAINING AGAINST THE SMART PARTIAL-BAND JAMMER (ONE REPRESENTATIVE SEED). TRAINING DIVERSIFIES THE POLICY AT $M=100$ BUT COLLAPSES IT ONTO ONE CHANNEL AT $M=1000$.

M	Updates	Avoid.	Distinct channels	Top-channel share
100	0	0.61	26	0.21
	220	0.53	19	0.31
	3,000	0.90	32	0.10
1,000	0	0.72	31	0.21
	220	0.04	17	0.93
	3,000	0.00	3	1.00

chosen action is updated, so each of the M output units needs its own samples. The exploration schedule spends about 15.5 random trials per channel in the first 3000 steps at $M=100$, but only 1.6 at $M=1000$. The few channels that happened to be rewarded early then dominate the argmax whatever the input. For the learned hopper, unpredictability is lost before competence is gained; for cryptographic hopping, unpredictability is a property of the key and cannot be trained away.



Fig. 3. Jamming-avoidance rate by strategy and jammer ($M=10$). DSSS is robust across every narrowband class and fails only against wideband barrage; fixed hopping fails broadly.

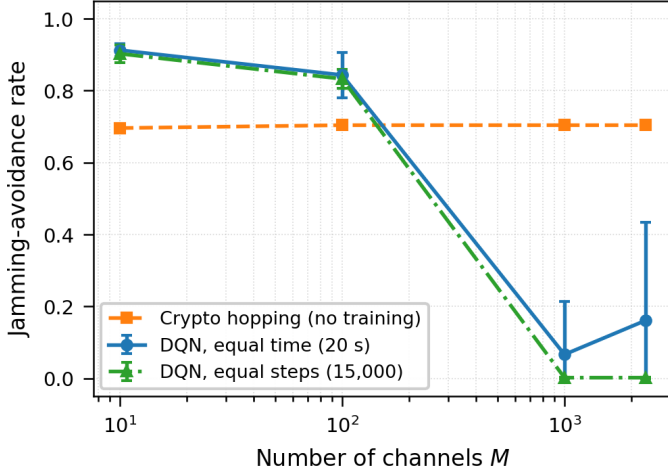


Fig. 4. Avoidance as channels scale, under both training regimes, against the training-free baseline. Points are means over seeds; bars are ± 1 standard deviation, clipped to the valid range $[0, 1]$.

D. Securing the position, and its cost

Table IV and Fig. 7 report attack-success rates over 2000 simulated attempts per layer. They check that each layer rejects the attack it is designed for under the requirements of Section IV-D, rather than measuring detection under noise: the simulation gives the drone the true geometric delay and a perfectly aligned clock. Cryptographic hopping alone reduces injection spoofing to $\approx 1/M$, because the spoofer must first guess the active channel, but leaves replay at 1.00. Adding a message signature stops spoofing, since a forged tag never verifies, yet a valid signature still loses to replay (1.00), because it does not protect arrival-time integrity. The full layer rejects both. In our replay model the attacker rebroadcasts a previously captured packet delayed by $20\sigma_r$ (11.7 m of range at 20 dB), so the stale nonce alone rejects it; a real-time rebroadcast would carry the current nonce and would be stopped only by the time-of-flight test.

Cost. Per decision, cryptographic hopping costs a single

TABLE IV
ATTACK-SUCCESS RATE BY SECURITY LAYER ($M=100$). A VALID SIGNATURE DOES NOT STOP REPLAY; AUTHENTICATED TIMING DOES, UNDER THE TIMING REQUIREMENTS OF SECTION IV-D.

Layer	Spoofing	Replay
None	1.00	1.00
Crypto hopping only	0.01	1.00
Crypto + signature only	0.00	1.00
Crypto + authenticated timing	0.00	0.00

AES block, an estimated 13 pJ on hardware AES, against 0.04–9.3 μJ for DL inference, a factor of up to 7.2×10^5 , and it requires no training (Fig. 8). The comparison is at the order-of-magnitude level: both figures are per-operation energy estimates rather than measurements on one platform.

Synchronization budget. We model de-hopping as failing once the clock offset between beacon and drone passes half the hop dwell, with Gaussian timing jitter of standard deviation 5% of the dwell. Holding de-hop success at 99% requires the offset to stay within about 0.37 of the dwell: 3.66 ms, 366 μs , and 36.6 μs at 100, 1000, and 10000 hops/s (Fig. 9). The budget shrinks in proportion to the hop rate. For a 1 ppm reference oscillator, which drifts 1 μs per second, these budgets are used up in roughly an hour, six minutes, and 37 s respectively, which sets how often the drone must resynchronize. Each resynchronization, an 8-slot late-entry acquisition in our model, costs 80, 8, and 0.8 ms.

E. Statistical validation

All comparisons use independently seeded runs. Differences are assessed with the paired t -test and the non-parametric Wilcoxon signed-rank test [36], with 95% bootstrap confidence intervals and Cohen's d effect size [37], at $\alpha=0.05$ with Bonferroni correction over the reported family. Table V confirms the crossover rather than asserting a single winner: the learned model significantly exceeds cryptographic hopping

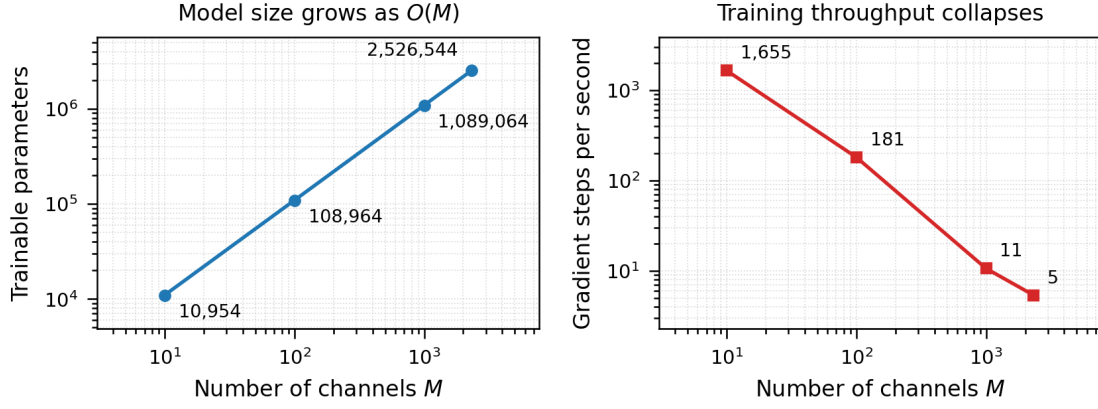


Fig. 5. $O(M)$ growth in parameters and the corresponding collapse in training throughput.

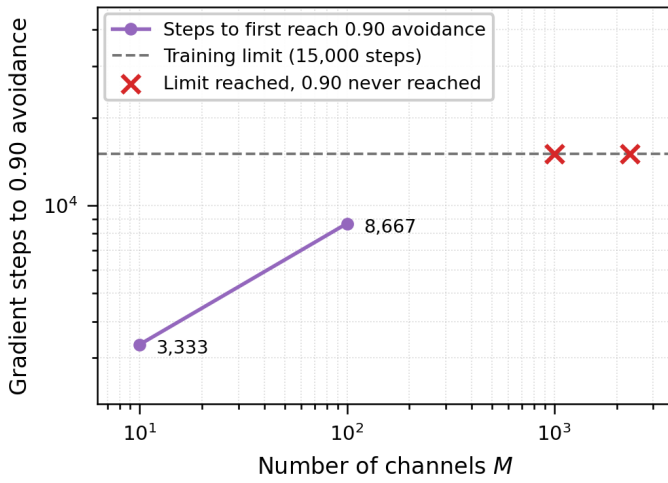


Fig. 6. Training cost of reaching a fixed competence level (0.90 avoidance) as the channel count grows.

at $M=10$ and $M=100$, and cryptographic hopping significantly exceeds it at $M=1000$. Against the intelligent learning jammer, cryptographic hopping exceeds the predictable public-random hopper by almost the full scale. Its differences from uncoordinated spread spectrum and DSSS are statistically detectable but below 0.01 in size; against that adversary all three unpredictable or spread schemes are effectively immune, and the separation that matters is between unpredictable and predictable strategies. The DQN rows come from the independent 8-seed replicate, trained for 8 s at $M=10$ and 20 s at $M=100$ and 1000, so their differences do not recompute exactly from Table II: at $M=1000$, for example, the replicate's DQN mean is 0.12 rather than 0.07.

VII. DISCUSSION

A. Boundary conditions

The findings are boundary-conditioned. A learned policy is the better anti-jammer *when* the channel count is small and training is affordable and *when* the adversary is not itself adaptive to the policy; it exploits structure that a blind keyed hopper ignores. The collapse analysis explains where that boundary comes from. A value-based learner has to try

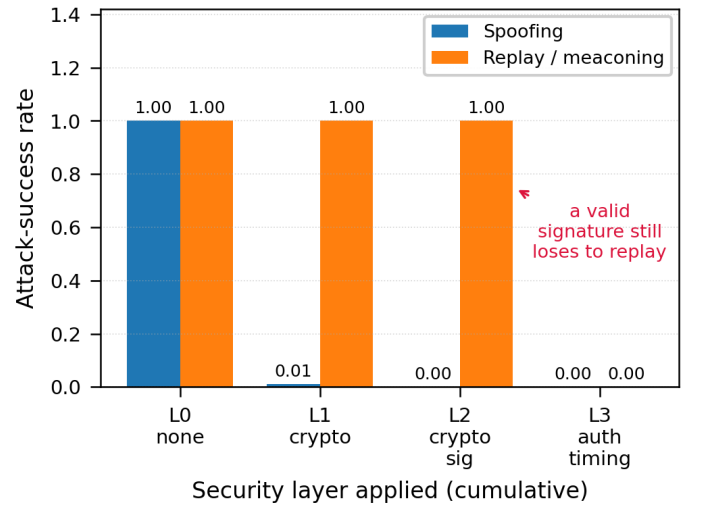


Fig. 7. Attack success by security layer ($M=100$). Layers are cumulative: L1 adds cryptographic hopping, L2 a message signature, and L3 authenticated timing. Spoofing injects forged beacon packets; replay re-transmits genuine packets with an added delay.

TABLE V
SIGNIFICANCE OF THE HEADLINE COMPARISONS, WITH p BONFERRONI-CORRECTED OVER THE SIX COMPARISONS. EACH ROW NAMES THE BETTER METHOD FIRST, SO EVERY DIFFERENCE IS POSITIVE. p IS FROM THE PAIRED t -TEST EXCEPT WHERE MARKED †: THERE EVERY PAIRED DIFFERENCE IS IDENTICAL, THE t STATISTIC IS UNDEFINED, AND p IS FROM THE WILCOXON SIGNED-RANK TEST; COHEN'S d IS LIKEWISE UNDEFINED (–).

Better > worse	Condition	Diff.	p	d
DQN > Crypto	$M=10$	0.215	$< 10^{-8}$	17.11
DQN > Crypto	$M=100$	0.155	$< 10^{-4}$	4.11
Crypto > DQN	$M=1000$	0.588	0.003	2.17
Crypto > Pub. random	learning	0.992	$< 10^{-6†}$	–
Crypto > USS	learning	0.003	0.034	0.55
DSSS > Crypto	learning	0.008	$< 10^{-6†}$	–

every channel enough times before it can distinguish them, and until it has, training does not make it neutral: it makes it predictable, which is the worst outcome against an adversary that tracks behavior. Cryptographic hopping wins *when* the band is wide, *when* training is impractical, and *when* the

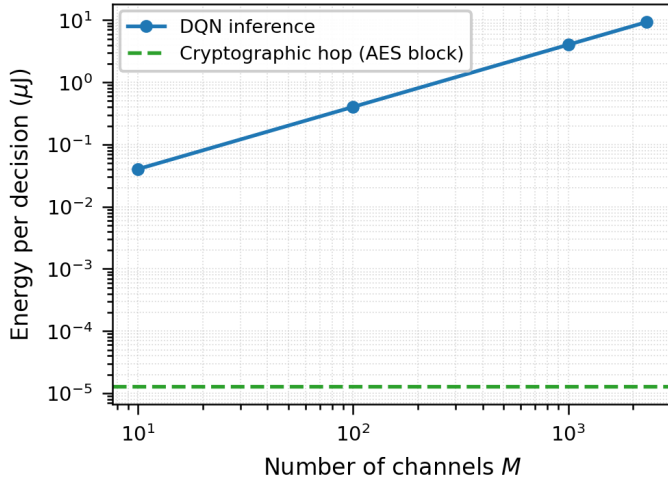


Fig. 8. Per-decision energy: cryptographic hop generation versus deep-learning inference.

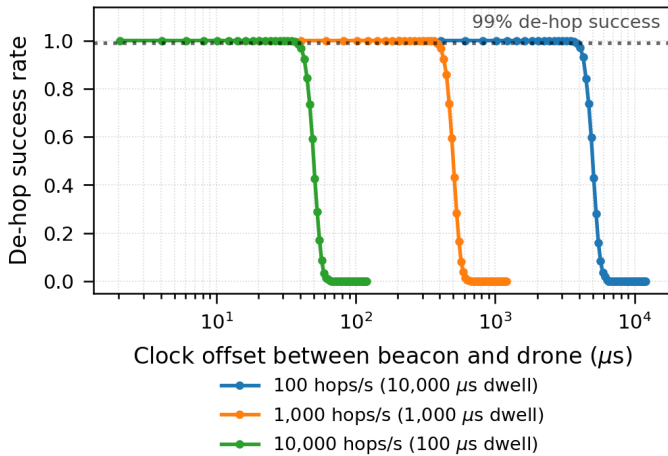


Fig. 9. De-hop success versus clock offset for three hop rates. The dotted line marks 99% success; each tenfold increase in hop rate cuts the tolerable offset tenfold.

adversary is intelligent, because unpredictability is a property of key secrecy, not of data. A wide band helps it against a jammer that covers a fixed number of channels, whose hit probability falls as $1/M$; against a jammer that covers a fixed fraction of the band, its collision rate stays at that fraction, which is the constant 0.70 in Table II. Only the cryptographic path additionally protects the *position*: with authenticated timing, and given the timing requirements of Section IV-D, it rejects jamming, spoofing, and replay together, which a communication-only learned hopper does not attempt.

B. Limits of the cryptographic approach

The cost analysis settles one common objection, that cryptography is as expensive as deep learning: generating a hop index is a single AES block, orders of magnitude below a neural forward pass and free of any training. The real costs are operational rather than computational, and they are not free. *Key distribution* is a logistical burden: keys must be provisioned before the mission and re-provisioned when

compromised, and the pre-shared assumption of Section IV-B is exactly what an adversary with physical access attacks. *Key secrecy is a single point of failure*: a disclosed key converts the unpredictable hopper into the predictable one, which the learning jammer drives to 0.00 avoidance. *Authenticated timing is demanding*: it needs nanosecond-level clock alignment and a range prediction finer than the TDOA fix itself (Section IV-D), and its nonce round makes the drone transmit. *Synchronization must be maintained*, and a drone that loses it pays the resynchronization latency of Section VI before it can receive at all. Finally, the cryptographic hopper is *blind*: it cannot exploit exploitable structure in the jammer, which is precisely why the learned model wins at small M . Neither approach answers wideband barrage, which defeats both.

C. Threats to validity

Construct validity. The jammer–hopper interaction is simulated at the channel-occupancy level, and range noise follows an analytic model that a signal-level GCC-PHAT check reproduces only near 20 dB; full RF propagation and hardware imperfections are abstracted. The security evaluation gives the drone the true geometric delay and a perfectly aligned clock, so it shows that the protocol logic is sound, not that a deployed drone meets those requirements. Two further consequences are worth stating plainly. The DSSS baseline is modeled by a processing-gain tolerance and therefore represents an upper bound on what a real DSSS receiver achieves. Conversely, the barrage jammer pays no penalty in this model for spreading its power across the whole band, whereas a real barrage adversary trades per-channel power density for coverage; the barrage column is therefore pessimistic for the defenders and should be read as a bound, not a prediction. *Internal validity*. The scaling result is reported under both a fixed wall-clock budget and a fixed number of gradient updates precisely because the first alone would confound the budget with the learning problem; the $O(M)$ growth in parameters and per-step cost is architectural and holds under either. The collapse diagnosis is drawn from one representative seed, and agrees with the zero variance of the equal-steps arm across all seeds at $M=1000$ and $M=2320$. The significance tests use an independent replicate whose DQN means differ slightly from those of the scaling study. It characterizes the reproduced learner, a value-based agent that updates only the chosen action under ϵ -greedy exploration. Learners that share information across actions, factorize the action space, or regularize their policy toward high entropy may resist collapse, and are the natural next comparison. *External validity*. Results are simulation-based with a single beacon geometry and a ten-class taxonomy; field behavior may differ, and hardware-in-the-loop validation remains future work. *Conclusion validity*. Multiple seeds, paired significance tests, confidence intervals, and Bonferroni correction guard against chance findings; NLOS metrics use medians because the error is heavy-tailed.

VIII. CONCLUSION

We benchmarked a reproduced deep-Q anti-jamming model against training-free cryptographic frequency hopping for

GPS-denied drone TDOA positioning, under one environment, dataset, and metric suite. The learned model matches or beats cryptographic hopping at small channel counts but fails as channels scale, and more training does not rescue it: at large channel counts training collapses its policy onto a single channel that an adaptive jammer tracks, so the learner loses its unpredictability before it gains competence. Cryptographic hopping stays effective, training-free, and unpredictable by construction, and, with an authenticated-timing layer, can also reject spoofed and replayed positioning signals, provided the drone can predict each beacon's time of flight to within nanoseconds, a requirement that sets the agenda for deployment. The maximum-likelihood TDOA estimator reaches the CRLB from 10 dB upward, and NLOS mitigation recovers most of the loss. Extending the taxonomy to wideband barrage bounds both families: it is the one adversary that defeats spread spectrum and hopping alike. We release the datasets, code, and benchmark. Future work extends the study to signal-level propagation, hardware-in-the-loop trials, a time-of-flight check driven by the drone's own tracking filter, adaptive-adversary co-training, and learners designed to resist policy collapse, such as entropy-regularized or action-factorized agents.

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